Fall 2024 CSCI 113-Lab 3; Noah Wiley

1. $CPU_{exe} = IC \cdot CPI \cdot CCT$ $\quad CCT = \frac{1}{\text{clock rate}}$

$CPI = \frac{CPU_{exe}}{IC} \cdot \text{clock rate}$ $\quad 800MHz \rightarrow GHz = 0.8 \cdot 10^9$ $\quad 700MHz \rightarrow GHz = 0.7 \times 10^9$

$M1 = (15s)(800\,MHz)/(600\,\text{million})$
$= 20$

$M2 = (20s)(700\,MHz)/(500\,\text{million})$
$= 28$

2. $IC = \frac{CPU_{exe}}{CPI \cdot CCT} = \frac{CPU_{exe} \cdot \text{clock rate}}{CPI}$

$M1 = (10s)(800\,MHz)/(20)$
$IC = 400$ million

$M2 = (8s)(700\,MHz)/(28)$
$IC = 200$ million

3. $MIPS = \frac{\text{Clock Rate}}{CPC}$

$M1 = \frac{800\,MHz}{1} = 800$ MIPS

$M2 = \frac{700\,MHz}{2} = 350$ MIPS

4. Average CPI for M_1 = $\frac{1+2+3+4}{4}$ = 2.5

Average CPI for M_2 = $\frac{2+3+4+2}{4}$ = 2.75

$$\text{MIPS} = \frac{\text{Clock Rate}}{\text{Average CPI}}$$

M_1 = 800 MHz / 2.5 = 320 MIPS
M_2 = 700 MHz / 2.75 = 254 MIPS

$\frac{320}{254}$ = 1.25

*The M_1 machine is 1.25 times faster than the M_2 machine

5. Effective MIPS = $\frac{\text{clock rate}}{\text{avg CPI}}$

The clock rate M_1 would need to have the same performance as M_2. We can set their peformance formulas equal.

$$\frac{M_1 \text{ Clock Rate}}{\text{CPI} = 2.5} = \frac{M_2 (700 \text{ MHz})}{2.75}$$

$$M_1 \text{ Clock Rate} = \frac{(700 \text{ MHz})(2.5)}{2.75}$$

M_1 Clock Rate = 636 MHz

6. $$\text{CPU time} = \frac{\text{CPU clock cycle}}{\text{Clock Rate}}$$

$$\frac{1}{2}\text{CPU time} = \frac{(1.4)(\text{clock cycles})}{\text{clock Rate}}$$

Clock Cycle = (CPU time)(Clock Rate)
= (20)(2.2 GHz) = 44

$$\text{Clock Rate}_{C_2} = \frac{(1.4)(\text{clock cycle})}{\text{CPU time}} \left(\frac{2}{1}\right)$$

$$= \frac{(1.4)(44)(2)}{20}$$

Clock Rate of C_2 = 6.16 GHz

7.

$$\frac{1}{(1-F_e)+\frac{F_e}{S_e}}$$

Speed up = 1.6

$F_e = 0.4$

$S_e = ?$

$$1 = \frac{\text{speedup}\left[(1-F_e)+\frac{F_e}{S_e}\right]}{\text{Speed up}} \rightarrow \frac{1}{\text{speedup}} = (1-F_e)+\frac{F_e}{S_e}-(1-F_e)$$

$$\frac{1}{\text{speedup}}-(1-F_e) = \frac{F_e}{S_e} = \frac{\frac{1}{\text{speedup}}-(1-F_e)}{F_e} = \frac{1}{S_e}$$

$$S_e = \frac{F_e}{(1/\text{speedup})-(1-F_e)}$$

$$S_e = \frac{0.4}{(1/1.6)-(1-0.4)} = \boxed{\text{16 units is the speedup needed.}}$$